dimensions anyway. However, if there were more dimensions of space, there would be more basis vectors. The dimension of a space can be defined as the maximum number of mutually orthogonal vectors in that space.

Obviously, there is nothing special about the particular axes *x*, *y*, and *z*. As long as the basis vectors are of unit length and are mutually orthogonal, they comprise an *orthonormal basis*.

The same principle is true for complex vector spaces. One can begin with any normalized vector and then look for a second one, orthogonal to the first. If you find one, then the space is at least two-dimensional. Then look for a third, fourth, and so on. Eventually, you may run out of new directions and there will not be any more orthogonal candidates. The maximum number of mutually orthogonal vectors is the dimension of the space. For column vectors, the dimension is simply the number of entries in the column.

Let's consider an *N*-dimensional space and a particular orthonormal basis of ket-vectors labeled $|i\rangle$.[7] The label *i* runs from 1 to *N*. Consider a vector $|A\rangle$, written as a sum of basis vectors:

$$|A\rangle = \sum_i \alpha_i |i\rangle. \tag{1.3}$$

The α_i are complex numbers called the *components* of the vector, and to calculate them we take the inner product of both sides with a basis bra $\langle j|$:

$$\langle j|A\rangle = \sum_i \alpha_i \langle j|i\rangle. \tag{1.4}$$

Next, we use the fact that the basis vectors are orthonormal. This implies that $\langle j|i\rangle = 0$ if *i* is not equal to *j*, and $\langle j|i\rangle = 1$ if $i = j$. In other words, $\langle j|i\rangle = \delta_{ij}$. This makes the sum in Eq. 1.4 collapse to one term:

$$\langle j|A\rangle = \alpha_j. \tag{1.5}$$